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Editor-in-Chief
Results in Engineering
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Dear Editor,

We are pleased to submit our manuscript entitled “**When Predictive Surrogates Fail as RL Environments: A Calibrated Physical Twin as Soft Regularizer for HVAC Control**” for consideration as a research article in *Results in Engineering*.

Why this work belongs in *Results in Engineering*. The journal’s scope explicitly includes reproducible engineering-software contributions in energy systems and building control, evaluated on standardised benchmarks. Our work addresses a real obstacle in deploying reinforcement-learning (RL) controllers to heating, ventilation and air-conditioning (HVAC) systems: standard RL training pipelines depend on neural-network surrogates whose predictive accuracy is routinely reported as a quality indicator, but whose usefulness as closed-loop training environments is rarely verified. We show that the two properties can decouple sharply, and we provide a reproducible recipe that resolves the resulting failure mode. All experiments use the open BOPTEST testbed; all code, configuration, trained models, and the eleven supplementary compliance tables (S1–S11) are released alongside the manuscript.

Novelty in four points.

1. **An explicit negative result with mechanistic explanation.** A physically calibrated RC-NeuralODE twin with identifiable zone thermal capacitance (C_{zon}) achieves a 0.64°C 24-hour rollout RMSE (versus 1.47°C uncalibrated) yet *collapses* as a stand-alone RL training environment, producing live closed-loop RMSE above 4°C . We trace this to a structural difference between predictive and closed-loop usage: residual errors that average out under one-step prediction accumulate into a positive-feedback loop under closed-loop training.
2. **A hybrid resolution.** We repurpose the calibrated twin as a frozen *soft physical regulariser* that enters the policy training loss but never the rollout forward pass. The resulting backend sustains 1,787 environment steps per second on a single CPU thread — an $85\times$ speed-up over the standard BOPTEST RTE HTTP–Docker deployment — and restores live closed-loop RMSE below 0.65°C .
3. **A controller-family-specific finding.** The optimal regularisation strength is not universal: $\lambda_{\text{temp}} = 0.10$ for thermostatic PPO but $\lambda_{\text{temp}} = 0.00$ for hierarchical RL and 17-dimensional MORL, with each value explained by observation-geometry and policy-hierarchy mechanisms. We report this as a falsification of the implicit “one λ fits all” assumption found in much of the prior literature.
4. **Pre-registered transferability decomposition.** Block 3 of the study cross-validates the recipe on three BOPTEST hydronic testcases under a manifest committed *before* any

non-source run. The surrogate component transfers structurally (identified C_{zon} ratio of 1.91 ± 0.03 across $N = 3$ testcases), while the frozen controller, routed through a mechanical hydronic adapter, fails on all three — with comfort-violation failure on the residential testcases and energy-inflation failure on the commercial one. This component-level decomposition of transfer is, to our knowledge, the first of its kind for surrogate-based HVAC RL.

Reproducibility and protocol compliance. The full pipeline follows the surrogate-development protocol of Hou & Evins (2024), with 15 of the 17 Reporting-Level-3 items documented quantitatively in Supplementary Tables S1–S11. Four git-anchored pre-registration commits (MORL canonical selection, MORL post- $N=5$ falsification, Block 3 pre-registration, Block 3 closure) bound the timeline against post-hoc adjustment.

Suggested reviewers. We suggest the following non-conflicted reviewers based on their recent contributions to RL-HVAC, surrogate modelling, or BOPTEST benchmarking. The authors have no personal, professional, or financial relationship with any of them.

- Davide Coraci (Politecnico di Torino) — staged RC calibration for building-energy surrogates.
- Yan Wang (Carnegie Mellon University) — integral safety metric for HVAC control benchmarking.
- David Blum (Lawrence Berkeley National Laboratory) — BOPTEST testbed maintainer and lead author.

We also welcome alternative reviewers chosen at the Editor’s discretion.

Originality and conflicts. The manuscript has not been published, has not been submitted elsewhere, and is not under review at any other journal. The author declares no competing financial interest or personal relationship that could be perceived as having influenced this work.

We thank you for your consideration and look forward to the reviewers’ comments.

Sincerely,

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